The Kepler Problem and Orbital Perturbation Theory

Contents

1 The Two-Body Problem

1.1 Central force problem

A central force acts along the line joining two particles and depends only on their separation $r = |\vec{x}_1 - \vec{x}_2|$. Such a force is a gradient of a scalar potential $V(r)$:

$$\vec{F}(\vec{x}_1, \vec{x}_2) = \nabla V(r).$$

Newton's equations for the pair are

$$\frac{d^2}{dt^2} \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{bmatrix} \vec{x}_1 \\ \vec{x}_2 \end{bmatrix} = \begin{bmatrix} -\nabla V(r) \\ \nabla V(r) \end{bmatrix}. \tag{1}$$

Six second-order ODEs $\Rightarrow$ 12 constants of integration in the general solution.

1.2 Separating centre-of-mass motion

With $M = m_1 + m_2$, $\mu = m_1 m_2 / M$, transform to relative and centre-of-mass coordinates,

$$\begin{bmatrix} \vec{r} \\ \vec{X} \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ m_1/M & m_2/M \end{bmatrix} \begin{bmatrix} \vec{x}_1 \\ \vec{x}_2 \end{bmatrix}, \qquad \begin{bmatrix} \vec{x}_1 \\ \vec{x}_2 \end{bmatrix} = \begin{bmatrix} m_2/M & 1 \\ -m_1/M & 1 \end{bmatrix} \begin{bmatrix} \vec{r} \\ \vec{X} \end{bmatrix}.$$

Substituting into (1) and left-multiplying by $\begin{bmatrix} m_2/M & -m_1/M \\ 1 & 1 \end{bmatrix}$ diagonalizes the mass matrix and decouples the two sectors:

$$\mu\,\ddot{\vec{r}} = -\nabla V(r), \tag{2}$$

$$M\,\ddot{\vec{X}} = 0. \tag{3}$$

Equation (3) integrates trivially: $\vec{X}(t) = (\vec{P}/M)t + \vec{X}_0$, with $\vec{P}, \vec{X}_0$ six constants fixing the inertial frame in which the barycentre is at rest at the origin. Everything of dynamical interest is in the *relative* coordinate $\vec{r}$, governed by (2) — a one-body central-force problem with reduced mass μ.

1.3 Conserved quantities

Angular momentum. Define $\vec{L} = \vec{r} \times \vec{p} = \mu\,\vec{r} \times \dot{\vec{r}}$. Then, because we are considering central force $\ddot{\vec{r}} \propto \hat{r}$ and hence,

$$\dot{\vec{L}} = \mu\big[\dot{\vec{r}} \times \dot{\vec{r}} + \vec{r} \times \ddot{\vec{r}}\big] = 0.$$

$\vec{L}$ supplies three constants of motion: its magnitude L, and two angles fixing its direction — the **inclination** ι and **longitude of ascending node** Ω (Euler angles of the orbital plane relative to a reference plane). Conservation of $\vec{L}$ confines the motion to the plane $\perp \vec{L}$.

Energy. $E = K + V$ with $K = \frac{1}{2}\mu\dot{\vec{r}}^2$ (the centre-of-mass kinetic term is a constant and drops out). Then

$$\dot{E} = \mu\dot{\vec{r}} \cdot \ddot{\vec{r}} + \dot{V} = \mu\dot{\vec{r}} \cdot \ddot{\vec{r}} + \nabla V \cdot \dot{\vec{r}} = \big(\mu\ddot{\vec{r}} + \nabla V\big) \cdot \dot{\vec{r}} = 0$$

by (2). So E is conserved for *any* central potential $V(r)$, not just gravity. Therefore, **total energy and angular momentum** constitutes four constant of integration. We will learn that there is one more, known as Runge Lenz vector which points towards periapsis and hence encodes the orientation of periapsis and the last constant of integration which is needed specifies the initial phase of the relative orbit (t_p)

$$\underbrace{\mathbf{R}_0, \mathbf{V}_{\mathrm{CM}}}_{6} + \underbrace{a, e, i, \Omega, \omega, t_p}_{6} = 12.$$

2 Coordinate Systems and Orbital Elements

Fix an equatorial (reference) frame with $\hat{\Upsilon}$ (vernal equinox / first point of Aries) along X and celestial north $\hat{N}$ along Z; $\hat{Y}$ completes the right-handed triad. For a source at $(\mathrm{RA}, \mathrm{DEC}) = (\alpha, \delta)$ the line of sight is

$$\hat{R} = \cos\delta\cos\alpha\,\hat{\Upsilon} + \cos\delta\sin\alpha\,\hat{Y} + \sin\delta\,\hat{N},$$

with in-sky-plane axes (pointing along increasing RA, DEC)

$$\hat{\alpha} = \frac{\frac{\partial\hat{R}}{\partial\alpha}}{\left|\frac{\partial\hat{R}}{\partial\alpha}\right|} = -\sin\alpha\,\hat{\Upsilon} + \cos\alpha\,\hat{Y}, \qquad \hat{\delta} = \frac{\frac{\partial\hat{R}}{\partial\delta}}{\left|\frac{\partial\hat{R}}{\partial\delta}\right|} = -\sin\delta\cos\alpha\,\hat{\Upsilon} - \sin\delta\sin\alpha\,\hat{Y} + \cos\delta\,\hat{N}.$$

The orbital plane has its own frame: origin at the barycentre, X-axis along the **line of nodes** (through the ascending node and the descending node), Z-axis along $\vec{L}$. Denote the unit vector towards the ascending node by $\hat{\Omega}$. Then

$$\begin{pmatrix} \hat{\Omega} \\ \hat{y} \\ \hat{L} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\iota & \sin\iota \\ 0 & -\sin\iota & \cos\iota \end{pmatrix} \begin{pmatrix} \cos\Omega & \sin\Omega & 0 \\ -\sin\Omega & \cos\Omega & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{\alpha} \\ \hat{\delta} \\ \hat{R} \end{pmatrix}. \tag{4}$$

Hence,

$$\hat{\Omega} = \cos\Omega\,\hat{\alpha} + \sin\Omega\,\hat{\delta}, \tag{5}$$

$$\hat{y} = -\sin\Omega\cos\iota\,\hat{\alpha} + \cos\Omega\cos\iota\,\hat{\delta} + \sin\iota\,\hat{R}, \tag{6}$$

$$\hat{L} = \sin\Omega\sin\iota\,\hat{\alpha} - \cos\Omega\sin\iota\,\hat{\delta} + \cos\iota\,\hat{R}, \tag{7}$$

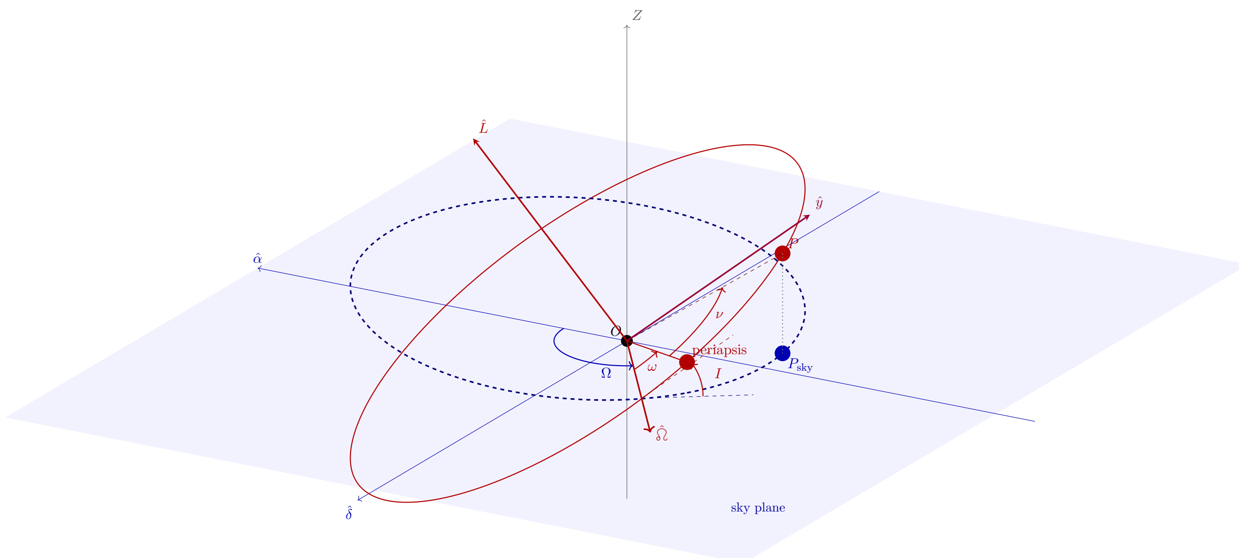


Figure 1: Orientation of an orbit relative to sky plane. Ω (longitude of ascending node) is measured from the reference direction α to the ascending node along the sky plane; ω (argument of periapsis) is measured from the ascending node to perigee *within* the orbital plane; ι (inclination) is the dihedral angle between the two planes, read off at the node; ν (true anomaly) is measured from perigee to the heavenly body's current position. This is exactly (Ω, ι, ω) of Eqs. (5)–(6) describe.

so that $\iota = \arccos\left(\hat{L}\cdot\hat{R}\right)$ is the angle between orbital and reference planes, and $(\hat{\Omega}, \hat{y}, \hat{L})$ is a right-handed orthonormal triad spanning the orbital plane plus its normal. In this plane use polar coordinates (r, ϕ) measured from $\hat{\Omega}$:

$$\hat{r} = \cos\phi\,\hat{\Omega} + \sin\phi\,\hat{y}, \qquad \hat{\phi} = -\sin\phi\,\hat{\Omega} + \cos\phi\,\hat{y}. \tag{8}$$

The projected orbit in the sky plane : If the object in its orbital plane as the coordinate $(r\cos(\omega+\nu), r\sin(\omega+\nu))$, the corresponding coordinate in the sky plane could be found as:

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix}_{\text{Sky Plane}} = \begin{pmatrix} \cos\Omega & -\sin\Omega & 0 \\ \sin\Omega & \cos\Omega & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos i & -\sin i \\ 0 & \sin i & \cos i \end{pmatrix} \begin{pmatrix} r\cos(\omega+\nu) \\ r\sin(\omega+\nu) \\ 0 \end{pmatrix}_{\text{Orbital Plane}} . \tag{9}$$

Then,

$$X = r\left(\cos(\omega+\nu)\cos\Omega - \sin(\omega+\nu)\sin\Omega\cos i\right), \tag{10}$$
$$Y = r\left(\cos(\omega+\nu)\sin\Omega + \sin(\omega+\nu)\cos\Omega\cos i\right). \tag{11}$$

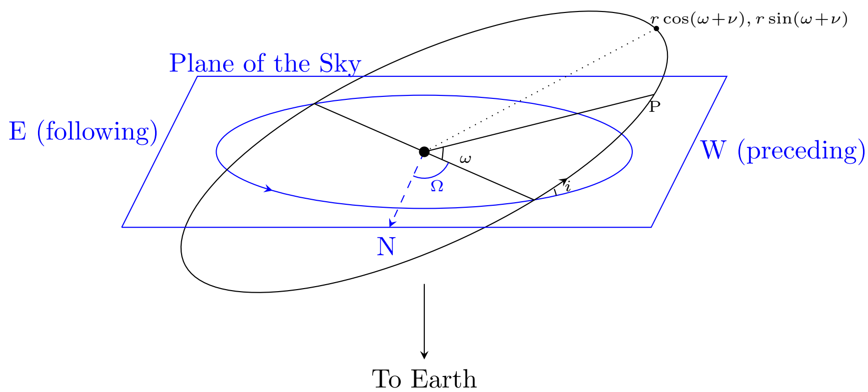


Figure 2: In this North becomes the local X axis and West becomes the Y axis of the sky plane.

One can set $\Omega = 0$ by another set of basis in the sky plane and then the analysis simplifies immensely

$$X = r\cos(\omega+\nu), \tag{12}$$
$$Y = r\sin(\omega+\nu)\cos i. \tag{13}$$

We find that the semi minor axis shrinks by the factor of $\cos i$.

The six orbital elements. As the orbit solution unfolds below, six constants emerge in total: Ω, ι (orientation of the plane, from $\vec{L}$), a, e (size and shape, from E, L), ω (orientation of the ellipse *within* the plane, the **argument of periapsis**), and T_0 (time of periapsis passage, fixing where on the ellipse the body is at a given time). Figure 1 shows all of these together.

3 Planar Motion and Kepler's Second Law

Areal velocity. The area swept by $\vec{r}$ in time dt is $dA = \int_0^{2\pi}\int_0^{r(\phi)} r dr d\phi = \frac{1}{2}\int_0^{2\pi} r^2(\phi) d\phi$, so

$$\frac{dA}{dt} = \frac{r^2\dot{\phi}}{2} = \frac{L}{2\mu} = \text{const.} \tag{14}$$

This is **Kepler's second law**: equal areas in equal times. It follows purely from $\vec{L}$ being conserved, i.e. from the force being central — no dependence on the specific form of $V(r)$.

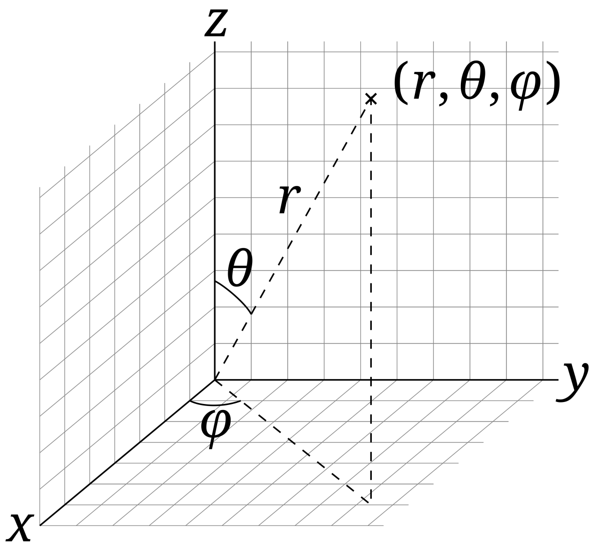


Figure 3: Schematic description of spherical polar coordinates.

Equations of motion in (r,ϕ)**.** With $Z \parallel \vec{L}$ and $X \parallel \hat{\Omega}$, $\vec{r}=r\hat{r}$, and

$$\dot{\vec{r}}=\dot{r}\,\hat{r}+r\dot{\hat{r}}=\dot{r}\,\hat{r}+r\dot{\phi}\,\hat{\phi}, \qquad \ddot{\vec{r}}=\left(\ddot{r}-r\dot{\phi}^2\right)\hat{r}+\left(2\dot{r}\dot{\phi}+r\ddot{\phi}\right)\hat{\phi}.$$

where we used $\nabla_\mu \mathbf{e}_\nu = \Gamma^\alpha_{\mu\nu}\mathbf{e}_\alpha$ which gives $\frac{\mathrm{d}\hat{r}}{\mathrm{d}t}=\frac{\partial\hat{r}}{\partial\phi}\dot{\phi}=\dot{\phi}\Gamma^\mu_{\phi r}\mathbf{e}_\mu=\dot{\phi}\frac{\mathbf{e}_\phi}{r}=\dot{\phi}\hat{\phi}$. Equation (2) splits into radial and tangential parts:

$$\mu\left(\ddot{r}-r\dot{\phi}^2\right)=-\nabla V(r), \tag{15}$$

$$\mu\left(2\dot{r}\dot{\phi}+r\ddot{\phi}\right)=0 \implies \frac{\mathrm{d}(r^2\dot{\phi})}{\mathrm{d}t}=0. \tag{16}$$

This was supposed to be a coupled differential equation. But because of equation (16) is just angular momentum conservation, i.e. Kepler's second law again, since $\vec{L}=\mu\vec{r}\times\dot{\vec{r}}=\mu r^2\dot{\phi}\,\hat{L}$. So

$$\dot{\phi}=\frac{L}{\mu r^2}, \tag{17}$$

and (15) becomes, eliminating $\dot{\phi}$,

$$\mu\ddot{r}-\frac{L^2}{\mu r^3}=-\nabla V(r). \tag{18}$$

4 The Orbit Equation

To get the orbit *shape* $r(\phi)$ we change independent variable from t to ϕ using (17): $\dfrac{dr}{dt}=\dot{\phi}\dfrac{dr}{d\phi}=\dfrac{L}{\mu r^2}\dfrac{dr}{d\phi}$, and

$$\frac{d^2r}{dt^2}=\frac{L}{\mu r^2}\frac{d}{d\phi}\left[\frac{L}{\mu r^2}\frac{dr}{d\phi}\right]=\frac{L^2}{\mu^2r^4}\left[\frac{d^2r}{d\phi^2}-\frac{2}{r}\left(\frac{dr}{d\phi}\right)^2\right].$$

Substituting into (18) and dividing by $L^2/(\mu r^4)$:

$$\frac{1}{r^4}\left[\frac{d^2r}{d\phi^2} - \frac{2}{r}\left(\frac{dr}{d\phi}\right)^2\right] - \frac{1}{r^3} = -\frac{\mu}{L^2}\nabla V(r).$$

The standard trick is the **Binet substitution** $r = 1/\rho$: then $dr/d\phi = -\rho^{-2}d\rho/d\phi$ and $d^2r/d\phi^2 = -\rho^{-2}d^2\rho/d\phi^2 + 2\rho^{-3}(d\rho/d\phi)^2$. Substituting and simplifying (the $(d\rho/d\phi)^2$ terms cancel), one obtains the **orbit equation**:

$$\boxed{\frac{d^2\rho}{d\phi^2} + \rho = \frac{\mu}{L^2}\frac{1}{\rho^2}\nabla V\left(\frac{1}{\rho}\right).} \tag{19}$$

This is a purely geometric ODE for the orbit's shape, valid for any central potential.

5 Keplerian Orbits

For Newtonian gravity, $V(r) = -GM\mu/r$, so $\nabla V(r) = GM\mu\rho^2\,\hat{r}$. Equation (19) becomes

$$\frac{d^2\rho}{d\phi^2} + \rho = \frac{GM\mu^2}{L^2}.$$

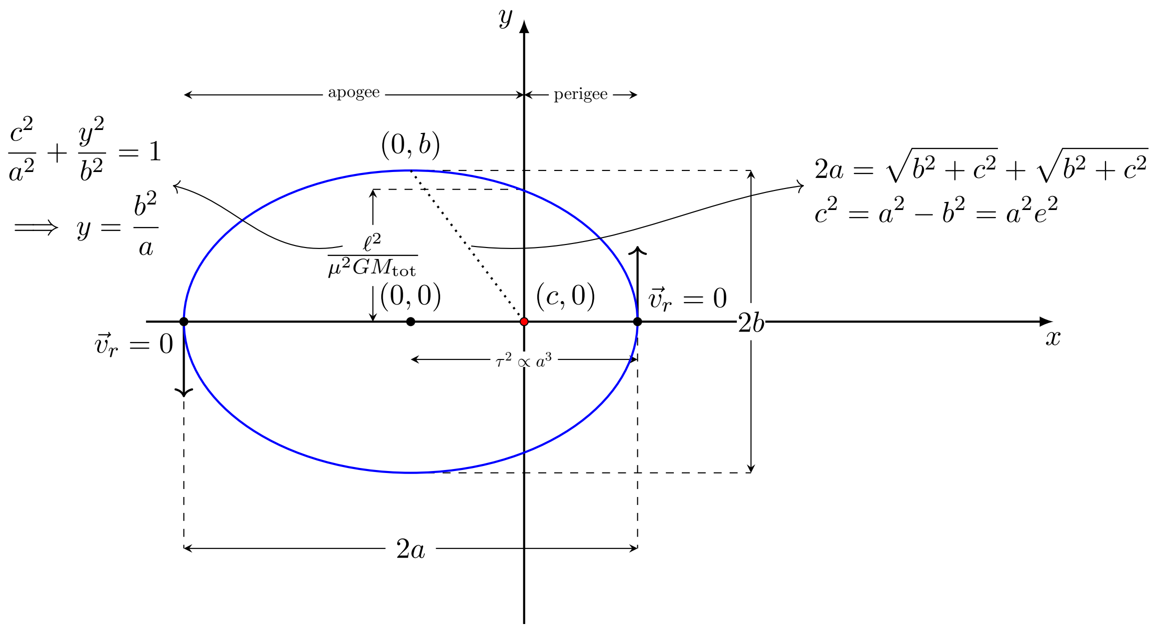
Shifting $\tilde{\rho} = \rho - GM\mu^2/L^2$ turns this into the simple-harmonic-oscillator equation $\ddot{\tilde{\rho}} + \tilde{\rho} = 0$ (derivatives in ϕ), with solution $\tilde{\rho} = A\cos(\phi - \omega)$ for constants A, ω. In terms of $r = 1/\rho$:

$$r(\phi) = \left(\frac{GM\mu^2}{L^2} + A\cos(\phi-\omega)\right)^{-1} = \frac{L^2/(GM\mu^2)}{1 + \dfrac{L^2A}{GM\mu^2}\cos(\phi-\omega)}.$$

Defining the **semi-latus rectum** $\lambda \equiv L^2/(GM\mu^2)$ as the value of r when $\phi = \omega + 90°$ and **eccentricity** $e \equiv \lambda A$,

$$\boxed{r(\phi) = \frac{\lambda}{1 + e\cos(\phi-\omega)}.} \tag{20}$$

This is the polar equation of a conic section with one focus at the origin, semi-latus rectum λ, eccentricity e, and major axis rotated by ω (the **argument of periapsis** — periapsis occurs at $\phi = \omega$).



Two more shape parameters follow:

$$a = \frac{\lambda}{1 - e^2} \text{ (semi-major axis)}, \qquad b = a\sqrt{1 - e^2} \text{ (semi-minor axis)}.$$

One can check this directly from the equation of the ellipse,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \tag{21}$$

by writing the polar coordinates about the focus as

$$x = ae + r(\phi)\cos\phi, \tag{22}$$
$$y = r(\phi)\sin\phi. \tag{23}$$

Substitution gives

$$\frac{(ae + r\cos\phi)^2}{a^2} + \frac{r^2 \sin^2\phi}{a^2(1 - e^2)} = 1. \tag{24}$$

Solving for r then yields

$$r(\phi) = \frac{a(1 - e^2)}{1 + e\cos\phi}. \tag{25}$$

Eccentricity	Conic section
$e = 0$	circle
$0 < e < 1$	ellipse ($E < 0$, bound)
$e = 1$	parabola ($E = 0$)
$e > 1$	hyperbola ($E > 0$, unbound)

Bound, closed elliptical orbits ($E < 0$) constitute **Kepler's first law**. In the original (unreduced) frame,

$$\vec{x}_1 = \frac{m_2}{M}\vec{r}, \qquad \vec{x}_2 = -\frac{m_1}{M}\vec{r}.$$

6 Energy and Angular Momentum in Terms of a, e

6.1 Energy

$$E = \frac{1}{2}\mu\dot{\vec{r}}^2 - \frac{GM\mu}{r} = \frac{1}{2}\mu\big(\dot{r}^2 + r^2\dot{\phi}^2\big) - \frac{GM\mu}{r} = \frac{1}{2}\mu\dot{r}^2 + \frac{L^2}{2\mu r^2} - \frac{GM\mu}{r}.$$

Since E is conserved, evaluate at periapsis ($\phi = \omega$), where $r = a - ae = \lambda/(1+e)$ and $\dot{r} = 0$ (turning point of r):

$$E = 0 + \frac{L^2(1+e)^2}{2\mu\lambda^2} - \frac{GM\mu(1+e)}{\lambda} = \frac{GM\mu^2\lambda(1+e)^2}{2\mu\lambda^2} - \frac{GM\mu(1+e)}{\lambda} = -\frac{GM\mu}{2\lambda}(1-e^2),$$

using $L^2 = GM\mu^2\lambda$ (below). With $a = \lambda/(1-e^2)$,

$$\boxed{E = -\frac{GM\mu}{2a}.} \tag{26}$$

Since $\lambda > 0$ always, the sign of E tracks e: bound orbits ($E < 0$) have $e < 1$, $a > 0$; unbound orbits ($E \geq 0$) have $e \geq 1$; hyperbolic orbits formally have $a < 0$.

6.2 Angular momentum

Both at apoapsis ($r = a + ae = \frac{\lambda}{1-e}$) and periapsis($r = a - ae = \frac{\lambda}{1+e}$), we have $\dot{r} = 0$,

$$\frac{L^2(1+e)^2}{2\mu\lambda^2} - \frac{GM\mu(1+e)}{\lambda} = \frac{L^2(1-e)^2}{2\mu\lambda^2} - \frac{GM\mu(1-e)}{\lambda} \tag{27}$$

Solving for L:

$$L = \sqrt{GM\mu^2\lambda}, \qquad \text{i.e.} \qquad L^2 = GM\mu^2 a(1-e^2), \tag{28}$$

and, using (7),

$$\vec{L} = \sqrt{GM\mu^2\lambda}\,\big(-\sin\Omega\sin\iota\,\hat{\alpha} + \cos\Omega\sin\iota\,\hat{\delta} + \cos\iota\,\hat{R}\big).$$

6.3 Instantaneous kinetic energy

From $E = K + V$: $K = \frac{1}{2}\mu\dot{\vec{r}}^2 = E - V = GM\mu\left(\frac{1}{r} - \frac{1}{2a}\right)$ — the *vis-viva* equation. Splitting between the two bodies via $\vec{x}_i$: $K_1 = \frac{1}{2}m_1 v_1^2 = \frac{m_2}{M}K$, $K_2 = \frac{m_1}{M}K$.

Inverting for a, e. Equations (26)–(28) give the orbital elements directly from the (conserved) energy and angular momentum:

$$a = -\frac{GM\mu}{2E}, \qquad e = \sqrt{1 + \frac{2EL^2}{(GM)^2\mu^3}}. \tag{29}$$

7 True and Eccentric Anomaly; Kepler's Equation

We now want $r(t)$ (and hence $\phi(t)$), not just $r(\phi)$. From the vis-viva relation,

$$\dot{r} = \sqrt{\frac{2}{\mu}\left(E - \frac{L^2}{2\mu r^2} + \frac{GM\mu}{r}\right)} = \sqrt{GM}\,\frac{1}{r}\left(2r - \frac{r^2}{a} - \lambda\right)^{1/2},$$

using (26)–(28) to trade E, L for a, λ. Then

$$\int_{t_0}^{t} dt = \frac{1}{\sqrt{GM}} \int_{r_0}^{r} \frac{r'\,dr'}{(2r' - r'^2/a - \lambda)^{1/2}} \tag{30}$$

$$t - t_0 = \sqrt{\frac{a}{GM}} \int_{r_0}^{r} \frac{r' dr'}{\sqrt{a^2e^2 - (r'-a)^2}} \qquad \text{(using } \lambda = a(1-e^2))$$

7.1 Elliptical orbits: eccentric anomaly

For $a > 0$, substitute

$$r = a\big(1 - e\cos u\big), \tag{31}$$

where u is the **eccentric anomaly** (this choice guarantees $r > 0$ throughout, since $-1 \le \cos u \le 1$). Then $dr = ae\sin u\,du$, and the bracket under the square root simplifies to $ae^2\sin^2 u$, so the square root is simply $\sqrt{a}\,e\,|\sin u|$. The integral collapses to

$$t - t_0 = \frac{a^{3/2}}{\sqrt{GM}} \int_0^u (1 - e\cos u')\,du' = \frac{a^{3/2}}{\sqrt{GM}}\big[u - e\sin u\big],$$

where t_0 (**epoch**) is the time at which $u = 0$ — the sixth and final constant of integration, T_0.

Relating true anomaly v and eccentric anomaly u: Define the **true anomaly** $v \equiv \phi - \omega$ (so $r = \lambda/(1 + e\cos v)$ from (20)). Equating this with (31):

$$\frac{1 - e\cos u}{1 - e^2} = \frac{1}{1 + e\cos v} \quad \Longrightarrow \quad \cos v = \frac{\cos u - e}{1 - e\cos u}. \tag{65a}$$

Consistently,

$$\sin v = \frac{\sqrt{1-e^2}\sin u}{1 - e\cos u}, \qquad \tan v = \frac{\sqrt{1-e^2}\sin u}{\cos u - e}, \qquad \tan\frac{v}{2} = \sqrt{\frac{1+e}{1-e}}\tan\frac{u}{2}. \tag{32}$$

At $u = 0$ and $u = \pi$, $v = u$ — periapsis and apoapsis coincide for the two anomalies, as they must by symmetry.

7.2 Hyperbolic trajectories

For $a < 0$ (and $e > 1$), the analogous substitution is $r = a(1 - e\cosh u)$ (using $\cosh u \ge 1$ keeps $r > 0$ since $a < 0$). Carrying through the same integral gives the **hyperbolic** anomaly relation

$$t - t_0 = \frac{(-a)^{3/2}}{\sqrt{GM}}\big[e\sinh u - u\big],$$

with true-anomaly relations

$$\cos v = \frac{e - \cosh u}{e\cosh u - 1}, \qquad \sin v = \frac{\sqrt{e^2-1}\sinh u}{e\cosh u - 1}, \qquad \tan\frac{v}{2} = \sqrt{\frac{e+1}{e-1}}\tanh\frac{u}{2}. \tag{33}$$

8 Kepler's Third Law

For one complete orbit, u advances by 2π while t advances by the orbital period $2\pi/n$ ($n \equiv$ mean motion):

$$\frac{2\pi}{n} = \frac{a^{3/2}}{\sqrt{GM}}\big[2\pi - e\sin(2\pi)\big] = \frac{a^{3/2}}{\sqrt{GM}} \cdot 2\pi \quad \Longrightarrow \quad \boxed{n = \sqrt{\frac{GM}{a^3}}, \qquad n^2a^3 = GM.} \tag{34}$$

9 Kepler's Equation

Substituting $n = \sqrt{GM/a^3}$ into $t - t_0 = (a^{3/2}/\sqrt{GM})(u - e\sin u)$ and defining the **mean anomaly** $\ell \equiv n(t - t_0)$:

$$\boxed{\ell = u - e\sin u} \qquad \text{(elliptical Kepler's equation).} \tag{35}$$

This transcendental equation must be solved numerically (or by series, §12) for $u(\ell)$, i.e. $u(t)$. The hyperbolic analogue (with $n \equiv GM/(-a)^{3/2}$) is

$$\ell = e\sinh u - u. \tag{36}$$

10 Keplerian Parametrization

10.1 Elliptical orbits

Collecting results:

$$r = a(1 - e\cos u), \qquad \phi = v + \omega, \tag{37}$$

$$\tan\frac{v}{2} = \sqrt{\frac{1-e}{1+e}}\tan\frac{u}{2}, \qquad \ell = u - e\sin u. \tag{38}$$

Differentiating Kepler's equation (35) in t: $n = \dot{u}(1 - e\cos u)$, so

$$\dot{u} = \frac{n}{1 - e\cos u}. \tag{39}$$

Differentiating $r = a(1 - e\cos u)$:

$$\dot{r} = ae\sin u\,\dot{u} = \frac{an\,e\sin u}{1 - e\cos u}. \tag{40}$$

Differentiating the v-u relation and using (32), one finds $dv/du = \sqrt{1-e^2}/(1 - e\cos u)$, so

$$\dot{\phi} = \dot{v} = \frac{dv}{du}\dot{u} = \frac{n\sqrt{1-e^2}}{(1 - e\cos u)^2}. \tag{41}$$

10.2 Hyperbolic trajectories

Analogously, with $r = -a(e\cosh u - 1)$, $\phi = v + \omega$, $\tan(v/2) = \sqrt{(e+1)/(e-1)}\tanh(u/2)$, $\ell = e\sinh u - u$:

$$\dot{u} = \frac{n}{e\cosh u - 1}, \quad \dot{r} = -an\frac{e\sinh u}{e\cosh u - 1}, \quad \dot{\phi} = \dot{v} = \frac{n\sqrt{e^2-1}}{(e\cosh u - 1)^2}. \tag{42}$$

11 Hyperbolic-Trajectory Parameters

For unbound ($e > 1$) encounters, four derived quantities are standard:

Hyperbolic excess velocity v_∞: from $\frac{1}{2}\mu\dot{\vec{r}}_\infty^2 = -GM\mu/(2a)$ (recall $a < 0$),

$$|\dot{\vec{r}}_\infty| = \sqrt{-\frac{GM}{a}}. \tag{43}$$

Scattering angle θ_∞: half the angle between the two asymptotes,

$$\cos\theta_\infty = -\lim_{u\to\infty}\frac{e-\cosh u}{e\cosh u - 1} = \frac{1}{e}. \tag{44}$$

Periapsis distance:

$$r_p = \min_v \frac{a(1-e^2)}{1+e\cos v} = -a(e-1). \tag{45}$$

Semi-minor axis / impact parameter: the semi-minor axis $b = -a\sqrt{e^2-1}$ satisfies $-ab = \lambda$. The impact parameter b' (miss distance of the unperturbed asymptote) is

$$b' = (-a + r_p)\sin\theta_\infty = -ae\sin\theta_\infty = -a\sqrt{e^2-1} = b,$$

i.e. impact parameter and semi-minor axis coincide.

Flight-path angle (angle between $\dot{\vec r}$ and the direction $\perp \hat r$ in-plane):

$$\sin\varphi = \frac{\dot{\vec r}\cdot\hat r}{|\dot{\vec r}|} = \frac{\dot r}{\sqrt{\dot r^2 + r^2\dot\phi^2}}. \tag{46}$$

12 Fourier Solution of Kepler's Equation

Kepler's equation $u - \ell = e\sin u$ can be solved as a Fourier series in ℓ:

$$e\sin u = \sum_{s=1}^{\infty} A_s \sin(s\ell).$$

Using $A_s = \frac{2}{\pi}\int_0^\pi (u-\ell)\sin(s\ell)\,d\ell$, integrating by parts (the boundary term vanishes since $u = \ell$ at $\ell = 0, \pi$), and substituting $d\ell = (1 - e\cos u)\,du$ (from Kepler's equation) turns the ℓ-integral into a u-integral:

$$A_s = \frac{2}{s\pi}\int_0^\pi \cos(su - se\sin u)\,du = \frac{2}{s}J_s(se),$$

recognizing the integral representation of the Bessel function of the first kind, $J_s(x) = \frac{1}{\pi}\int_0^\pi \cos(s\tau - x\sin\tau)\,d\tau$. So

$$u(\ell) = \ell + \sum_{s=1}^{\infty}\frac{2}{s}J_s(se)\sin(s\ell). \tag{47}$$

This is the classical Bessel-function series solution to Kepler's equation.

13 The Laplace–Runge–Lenz Vector

Define $\vec A \equiv \vec p \times \vec L - GM\mu^2\hat r$ (with $\vec p = \mu\dot{\vec r}$). For the pure Kepler problem,

$$\dot{\vec A} = \dot{\vec p}\times\vec L - GM\mu^2\dot{\hat r} = -\frac{GM\mu}{r^2}\big(\hat r\times\vec L\big) - GM\mu^2\dot\phi\,\hat\phi.$$

Using $\vec L = L\hat L$, planar motion identities $\hat r \times \hat L = -\hat\phi$, and $\dot\phi = L/(\mu r^2)$ from (17):

$$\dot{\vec A} = -GM\mu\left(\frac{L}{r^2}\hat\phi - \mu\dot\phi\,\hat\phi\right) = 0,$$

so $\vec A$ is conserved for the unperturbed problem — it is a *fourth*, non-obvious constant of the Kepler problem (beyond $E, \vec L$), reflecting the extra "hidden" $SO(4)$ symmetry of the $1/r$ potential.

Direction and magnitude of $\vec{A}$. From the definition it is clear that $\vec{A}$ is planar. Therefore, we can decompose it as $\vec{A} = A_r\hat{r} + A_\phi\hat{\phi}$.

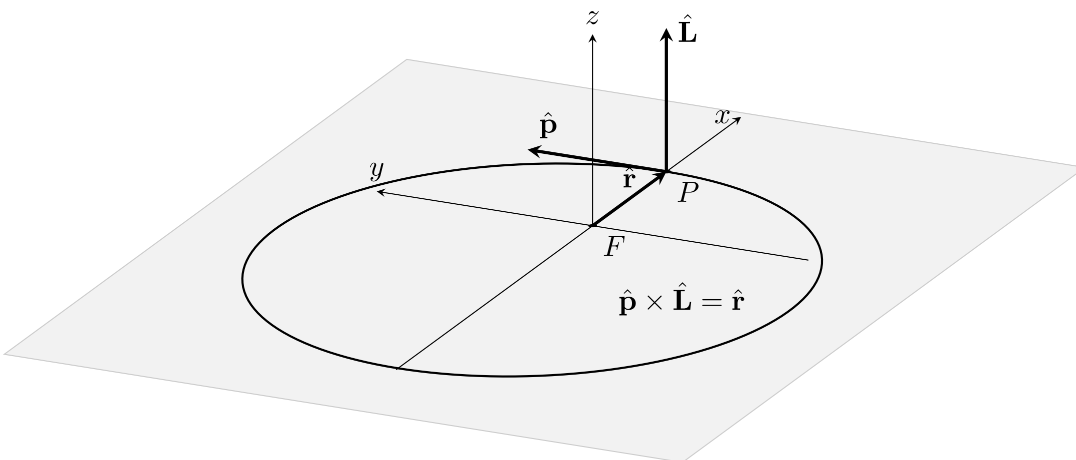


Figure 4: The Runge Lenz vector points radially outward at perigee. Since it is conserved, the direction and magnitude stays fixed during the entire evolution.

At periapsis the velocity is purely tangential ($\vec{p} \perp \hat{r}$), so $\hat{r} \times \vec{A}\big|_{\text{peri}} = \hat{r} \times (\vec{p} \times \vec{L})\big|_{\text{peri}} = \big[\vec{p}(\hat{r} \cdot \vec{L}) - \vec{L}(\hat{r} \cdot \vec{p})\big]_{\text{peri}} = 0$ (both dot products vanish there). Also $\hat{r} \cdot \vec{A}|_{\text{peri}} = A$ (from the relation above with $v = 0$ at periapsis). Together these mean $\hat{A} = \hat{r}|_{\text{peri}}$: $\vec{A}$ points from the focus *towards periapsis.*

Since $\vec{A}$ points radially outwards at the Periapsis. We find the magnitude by doting $\vec{A}$ into $\vec{r}$:

$$\vec{A} \cdot \vec{r} = (\vec{p} \times \vec{L}) \cdot \vec{r} - GM\mu^2 r \tag{48}$$
$$= p_i L_j \epsilon_{ijk} r_k - GM\mu^2 r = p_i L_j \epsilon_{kij} r_k - GM\mu^2 r \tag{49}$$
$$= \underbrace{(\vec{r} \times \vec{p})}_{\vec{L}} \cdot \vec{L} - GM\mu^2 r = L^2 - GM\mu^2 r, \tag{50}$$

i.e., writing $\vec{A} \cdot \vec{r} = Ar\cos v$ (with v the angle between $\vec{A}$ and $\vec{r}$), using

$$r = \frac{L^2/(GM\mu^2)}{1 + e\cos v} \implies GM\mu^2 r = \frac{L^2}{1 + e\cos v}.$$

We find:

$$Ar\cos v = L^2 - GM\mu^2 r, \tag{51}$$
$$A\frac{L^2/GM\mu^2}{1 + e\cos v}\cos v = L^2\left(1 - \frac{1}{1 + e\cos v}\right) = \frac{L^2 e\cos v}{1 + e\cos v} \tag{52}$$

Hence

$$\boxed{A = GM\mu^2 e.} \tag{53}$$

Using (8):

$$\hat{A} = \cos\omega\,\hat{\Omega} + \sin\omega\,\hat{y}. \tag{54}$$

Equivalently, the **eccentricity vector** $\vec{e} \equiv \vec{A}/(GM\mu^2) = e\hat{A} = e(\cos\omega, \sin\omega, 0)$ is a conserved vector encoding (e, ω) together. Since e, Ω, ι are already determined by $(E, \vec{L})$, the only *new* information in $\vec{A}$ is ω, i.e. the in-plane orientation of the ellipse or the direction of periapsis from the line of ascending node.

14 Perturbed Orbits

We now allow an additional force $\vec{F}$ in the relative equation of motion:

$$\boxed{\mu\ddot{\vec{r}} = -\frac{GM\mu}{r^2}\hat{r} + \vec{F}.} \tag{55}$$

The central Newtonian term defines the Kepler problem; $\vec{F}$ is the perturbing force and need not be central.

The momentum is

$$\vec{p} = \mu\dot{\vec{r}}, \tag{56}$$

and, in the orbital plane,

$$\vec{p} = \mu\dot{r}\,\hat{r} + \frac{L}{r}\,\hat{\phi}, \qquad L = \mu r^2\dot{\theta}. \tag{57}$$

The purpose of orbital perturbation theory is to represent the actual trajectory by Keplerian orbital elements which vary with time, and to derive their rates directly from (55).

14.1 Osculating Kepler orbit

For $\vec{F} = 0$, the Kepler solution can be written in terms of six constants

$$Q = (Q_1, \ldots, Q_6) = (a, e, \iota, \Omega, \omega, T_0), \tag{58}$$

so that

$$\vec{r} = \vec{r}(Q, t), \qquad \vec{p} = \vec{p}(Q, t). \tag{59}$$

For fixed Q, these functions satisfy the unperturbed equations of motion.

For the perturbed problem, let

$$Q_i = Q_i(t). \tag{60}$$

The actual trajectory is then represented in the form

$$\vec{r}(t) = \vec{r}(Q(t), t), \qquad \vec{p}(t) = \vec{p}(Q(t), t). \tag{61}$$

This representation does not assert that the actual trajectory is a Keplerian trajectory. At each instant it defines a Kepler orbit which has the same position and momentum as the actual trajectory at that instant. Thus

$$\vec{r}_{\rm osc}(t) = \vec{r}(t), \qquad \vec{p}_{\rm osc}(t) = \vec{p}(t). \tag{62}$$

The corresponding accelerations are different:

$$\dot{\vec{p}}_{\rm actual} = -\frac{GM\mu}{r^2}\hat{r} + \vec{F}, \tag{63}$$

$$\dot{\vec{p}}_{\rm osc} = -\frac{GM\mu}{r^2}\hat{r}. \tag{64}$$

Hence

$$\boxed{\dot{\vec{p}}_{\text{actual}} - \dot{\vec{p}}_{\text{osc}} = \vec{F}.} \tag{65}$$

The two trajectories therefore agree in position and momentum at the instant at which the osculating orbit is constructed, but they have different subsequent evolution.

14.2 Variation of parameters

Differentiating (61),

$$\dot{\vec{r}} = \left.\frac{\partial \vec{r}}{\partial t}\right|_Q + \sum_{i=1}^{6} \frac{\partial \vec{r}}{\partial Q_i}\dot{Q}_i. \tag{66}$$

For fixed Q, the first term is the velocity of the Kepler solution. The osculating condition requires this Kepler velocity to coincide with the actual velocity:

$$\dot{\vec{r}} = \left.\frac{\partial \vec{r}}{\partial t}\right|_Q. \tag{67}$$

Therefore

$$\boxed{\sum_{i=1}^{6} \frac{\partial \vec{r}}{\partial Q_i}\dot{Q}_i = 0.} \tag{68}$$

This is a vector equation and therefore supplies three scalar conditions.

Now differentiate $\vec{p}(Q(t), t)$:

$$\dot{\vec{p}} = \left.\frac{\partial \vec{p}}{\partial t}\right|_Q + \sum_{i=1}^{6} \frac{\partial \vec{p}}{\partial Q_i}\dot{Q}_i. \tag{69}$$

For fixed Q, $\vec{p}(Q, t)$ obeys the Kepler equation,

$$\left.\frac{\partial \vec{p}}{\partial t}\right|_Q = -\frac{GM\mu}{r^2}\hat{r}. \tag{70}$$

The actual momentum obeys

$$\dot{\vec{p}} = -\frac{GM\mu}{r^2}\hat{r} + \vec{F}. \tag{71}$$

Subtracting the Kepler contribution gives

$$\boxed{\sum_{i=1}^{6} \frac{\partial \vec{p}}{\partial Q_i}\dot{Q}_i = \vec{F}.} \tag{72}$$

Equations (68) and (72) form six scalar equations for the six quantities $\dot{Q}_i$. This is the formal variation-of-parameters construction of the osculating elements.

The perturbing force therefore enters through the change in the instantaneous phase-space state; the Kepler orbit is redefined at each instant from that state.

14.3 Perturbed Kepler invariants

The Keplerian energy, angular momentum and eccentricity vector are defined from the instantaneous state by

$$E \equiv \frac{p^2}{2\mu} - \frac{GM\mu}{r}, \tag{73}$$

$$\vec{L} \equiv \vec{r} \times \vec{p}, \tag{74}$$

$$\vec{A} \equiv \frac{\vec{p} \times \vec{L}}{\mu} - GM\mu\,\hat{r}. \tag{75}$$

For the unperturbed Kepler problem these are constants. In the perturbed problem they become time-dependent quantities.

Their magnitudes determine the instantaneous Keplerian parameters:

$$E = -\frac{GM\mu}{2a(t)}, \tag{76}$$

$$L^2 = GM\mu^2 a(t)(1 - e^2(t)), \tag{77}$$

$$A \equiv |\vec{A}| = GM\mu^2 e(t). \tag{78}$$

Thus a and e may be regarded as functions of the actual state even when the full trajectory is not Keplerian.

14.4 Exact Rate of the Energy

Differentiate (73):

$$\dot{E} = \frac{\vec{p}}{\mu} \cdot \dot{\vec{p}} + \frac{GM\mu}{r^2}\dot{r}. \tag{79}$$

Using

$$\dot{\vec{p}} = -\frac{GM\mu}{r^2}\hat{r} + \vec{F} \tag{80}$$

and

$$\frac{\vec{p}}{\mu} \cdot \hat{r} = \dot{r}, \tag{81}$$

we obtain

$$\dot{E} = \dot{\vec{r}} \cdot \left(-\frac{GM\mu}{r^2}\hat{r} + \vec{F}\right) + \frac{GM\mu}{r^2}\dot{r} \tag{82}$$

$$= -\frac{GM\mu}{r^2}\dot{r} + \dot{\vec{r}} \cdot \vec{F} + \frac{GM\mu}{r^2}\dot{r} \tag{83}$$

$$= \boxed{\dot{E} = \dot{\vec{r}} \cdot \vec{F}.} \tag{84}$$

14.5 Exact Rate of angular momentum

From

$$\vec{L} = \vec{r} \times \vec{p}, \tag{85}$$

$$\dot{\vec{L}} = \dot{\vec{r}} \times \vec{p} + \vec{r} \times \dot{\vec{p}}. \tag{86}$$

Since $\vec{p} = \mu\dot{\vec{r}}$,

$$\dot{\vec{r}} \times \vec{p} = 0, \tag{87}$$

and hence

$$\dot{\vec{L}} = \vec{r} \times \left(-\frac{GM\mu}{r^2}\hat{r} + \vec{F}\right). \tag{88}$$

The central term vanishes because $\vec{r} \parallel \hat{r}$, giving

$$\boxed{\dot{\vec{L}} = \vec{r} \times \vec{F}.} \tag{89}$$

Thus only the component of $\vec{F}$ perpendicular to $\vec{r}$ produces torque.

14.6 Exact Rate of the eccentricity vector

Differentiate

$$\vec{A} = \vec{p} \times \vec{L} - GM\mu^2\,\hat{r} : \tag{90}$$

$$\dot{\vec{A}} = \dot{\vec{p}} \times \vec{L} + \vec{p} \times \dot{\vec{L}} - GM\mu^2\,\dot{\hat{r}}. \tag{91}$$

Insert

$$\dot{\vec{p}} = -\frac{GM\mu}{r^2}\hat{r} + \vec{F} \tag{92}$$

and

$$\dot{\vec{L}} = \vec{r} \times \vec{F}. \tag{93}$$

Then

$$\dot{\vec{A}} = -\frac{GM\mu}{r^2}\hat{r} \times \vec{L} + \vec{F} \times \vec{L} + \vec{p} \times (\vec{r} \times \vec{F}) - GM\mu^2\,\dot{\hat{r}}. \tag{94}$$

For the Kepler part,

$$\dot{\hat{r}} = \frac{\vec{L} \times \vec{r}}{\mu r^3}, \tag{95}$$

so

$$-GM\mu^2\,\dot{\hat{r}} = -\frac{GM\mu}{r^2}\left(\vec{L} \times \hat{r}\right) \tag{96}$$

$$= \frac{GM\mu}{r^2}\,\hat{r} \times \vec{L}. \tag{97}$$

The two central terms cancel. Therefore

$$\boxed{\dot{\vec{A}} = \vec{F} \times \vec{L} + \vec{p} \times (\vec{r} \times \vec{F}).} \tag{98}$$

This equation is the starting point for the perturbative evolution of the eccentricity and the orientation of periapsis.

15 Planar perturbations

We now specialize to a perturbing force confined to the orbital plane:

$$\boxed{\vec{F} = F_r \hat{r} + F_\theta \hat{\phi}.} \tag{99}$$

The angular momentum vector is perpendicular to the plane,

$$\vec{L} = L\hat{L}. \tag{100}$$

Consequently the plane itself remains fixed:

$$\boxed{i = 0, \qquad \dot{\Omega} = 0.} \tag{101}$$

In the orbital plane,

$$\vec{p} = \mu\dot{r}\,\hat{r} + \mu r\dot{\theta}\,\hat{\phi} = \mu\dot{r}\,\hat{r} + \frac{L}{r}\hat{\phi}. \tag{102}$$

The equations of motion follow directly from (55):

$$\mu(\ddot{r} - r\dot{\theta}^2) = -\frac{GM\mu}{r^2} + F_r, \tag{103}$$

$$\mu(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = F_\theta. \tag{104}$$

The second equation gives

$$\frac{dL}{dt} = \frac{d}{dt}(\mu r^2\dot{\theta}) = rF_\theta, \tag{105}$$

so

$$\boxed{\dot{L} = rF_\theta.} \tag{106}$$

The energy equation becomes

$$\dot{E} = \dot{r}F_r + r\dot{\theta}F_\theta \tag{107}$$

$$= \boxed{\dot{r}F_r + \frac{L}{\mu r}F_\theta.} \tag{108}$$

15.1 Keplerian geometry of the instantaneous ellipse

Define the true anomaly

$$f \equiv \theta - \omega. \tag{109}$$

For the instantaneous Kepler orbit,

$$\boxed{r = \frac{\lambda}{1 + e\cos f},} \tag{110}$$

where

$$\boxed{\lambda \equiv a(1 - e^2) = \frac{L^2}{GM\mu^2}.} \tag{111}$$

The instantaneous Keplerian angular velocity is therefore

$$\dot{\theta} = \frac{L}{\mu r^2}. \tag{112}$$

We can find the instantaneous radial velocity of the osculating Keplerian orbit from (110). In general, for an evolving orbit,

$$\dot{r} = \frac{\partial r}{\partial t} + \frac{\partial r}{\partial a}\dot{a} + \frac{\partial r}{\partial e}\dot{e} + \frac{\partial r}{\partial f}\dot{f}. \tag{113}$$

For the osculating Keplerian orbit at a given instant, we fix a and e at their instantaneous values and consider only the Keplerian motion in f. Thus,

$$\boxed{\dot{r} = \frac{dr}{df}\dot{f} = \frac{\lambda e \sin f}{(1+e\cos f)^2}\frac{L}{\mu r^2} = \frac{Le\sin f}{\mu\lambda}.} \tag{114}$$

Equivalently, this is the instantaneous radial velocity obtained by removing the perturbing force at that instant while keeping the instantaneous $\vec{r}$ and $\vec{p}$ unchanged.

Since

$$\frac{\lambda}{r} = 1 + e\cos f, \tag{115}$$

we also have

$$\boxed{\frac{\mu r\dot{r}}{L} = \frac{e\sin f}{1+e\cos f}.} \tag{116}$$

Define

$$\boxed{\xi \equiv \frac{e\sin f}{1+e\cos f}.} \tag{117}$$

15.2 Rate of the semi-major axis

At this particular instant, we are asking if there exists a Kepler ellipse whose energy is equal to the instantaneous energy of the real orbit given by (84). Let's label that ellipse by $a(t)$ and it will have energy given by:

$$E\Big|_{\text{osculating}} = -\frac{GM\mu}{2a} \implies \dot{E}\Big|_{\text{osculating}} = \frac{GM\mu}{2a^2}\dot{a}. \tag{118}$$

Therefore, we can replace the distance r and radial velocity $\dot{r}$ on the right-hand side of (108) with their corresponding values for the osculating Keplerian orbit:

$$\dot{E}\Big|_{\text{osculating}} = \dot{r}F_r + \frac{L}{\mu r}F_\theta\Big|_{\text{osculating}} = \frac{L}{\mu\lambda}e\sin f\, F_r + \frac{L}{\mu r}F_\theta. \tag{119}$$

Hence we find:

$$\dot{a} = \frac{2a^2 L}{GM\mu^2}\left[\frac{e\sin f}{\lambda}F_r + \frac{1}{r}F_\theta\right]. \tag{120}$$

Since

$$L = \mu\sqrt{GM\lambda}, \qquad \lambda = a(1-e^2), \tag{121}$$

this becomes

$$\boxed{\dot{a} = \sqrt{\frac{4a^3}{GM\mu^2(1-e^2)}}\left[e\sin f\, F_r + (1+e\cos f)F_\theta\right].} \tag{122}$$

15.3 Rate of the eccentricity

Since

$$A = GM\mu^2 e, \tag{123}$$

it is sufficient to calculate $\dot{A}$ from (98). For the planar perturbation, we use

$$\vec{L} = L\hat{L} \tag{124}$$

together with

$$\begin{aligned} \vec{F} \times \vec{L} &= L\left(F_r\hat{r} + F_\theta\hat{\phi}\right) \times \hat{L} && (125)\\ &= L\left(-F_r\hat{\phi} + F_\theta\hat{r}\right). && (126)\\ \vec{p} \times (\vec{r} \times \vec{F}) &= \left(\mu\dot{r}, \hat{r} + \frac{L}{r}\hat{\phi}\right) \times \left(rF_\theta\hat{L}\right) && (127)\\ &= LF_\theta\hat{r} - \mu r\dot{r}F_\theta\hat{\phi}. && (128) \end{aligned}$$

Hence

$$\dot{\vec{A}} = \vec{F} \times \vec{L} + \vec{p} \times (\vec{r} \times \vec{F}) = 2LF_\theta\hat{r} - (LF_r + \mu r\dot{r}F_\theta)\,\hat{\phi}. \tag{129}$$

Using (117),

$$\boxed{\dot{\vec{A}} = L\left[2F_\theta\hat{r} - (F_r + \xi F_\theta)\hat{\phi}\right].} \tag{130}$$

The unit vector in the direction of $\vec{A}$ is the direction of periapsis. Since the true anomaly is measured from periapsis,

$$\boxed{\hat{A} = \cos f\,\hat{r} - \sin f\,\hat{\phi}.} \tag{131}$$

Notice,

$$\dot{\vec{A}} = \dot{A}\hat{A} + A\dot{\hat{A}} \tag{132}$$

$$\dot{\vec{A}} \cdot \hat{A} = \dot{A} \qquad \left(\text{using } \frac{\mathrm{d}\hat{A}\cdot\hat{A}}{\mathrm{d}t} = 0\right)$$

Therefore

$$\begin{aligned} \dot{A} &= \dot{\vec{A}} \cdot \hat{A} && (133)\\ &= L\left[2F_\theta \cos f + (F_r + \xi F_\theta)\sin f\right]. && (134) \end{aligned}$$

Thus

$$\boxed{\dot{A} = L\left[F_r \sin f + F_\theta\left(2\cos f + \xi \sin f\right)\right].} \tag{135}$$

Up to this point, all relations have been exact. We now evaluate the exact rate equation using the instantaneous osculating Keplerian orbit, replacing the corresponding quantities on both the left- and right-hand sides by their osculating values.

$$\dot{A}\Big|_{\text{osculating}} = L\left[F_r \sin f + F_\theta\left(2\cos f + \xi \sin f\right)\right]\Big|_{\text{osculating}} \tag{136}$$

$$GM\mu^2\dot{e} = \sqrt{GM\mu^2 a(1-e^2)}\left[F_r \sin f + F_\theta\left(2\cos f + \xi \sin f\right)\right] \tag{137}$$

we obtain

$$\boxed{\dot{e} = \sqrt{\frac{a(1-e^2)}{GM\mu^2}}\left[F_r \sin f + F_\theta\left(2\cos f + \xi \sin f\right)\right].} \tag{138}$$

15.4 Rate of periapsis

Define the unit vector perpendicular to $\hat{A}$ in the direction of increasing true anomaly:

$$\boxed{\hat{q} = \sin f\, \hat{r} + \cos f\, \hat{\phi}.} \tag{139}$$

Differentiating (131) gives

$$\dot{\hat{A}} = -\dot{f} \sin f\, \hat{r} + \cos f\, \dot{\hat{r}} - \dot{f} \cos f\, \hat{\phi} - \sin f\, \dot{\hat{\phi}}, \tag{140}$$

with

$$\dot{\hat{r}} = \dot{\theta}\, \hat{\phi}, \qquad \dot{\hat{\phi}} = -\dot{\theta}\, \hat{r}. \tag{141}$$

Thus

$$\dot{\hat{A}} = (\dot{\theta} - \dot{f}) \left(\sin f\, \hat{r} + \cos f\, \hat{\phi}\right) \tag{142}$$

$$= \dot{\omega}\, \hat{q}, \tag{143}$$

because

$$f = \theta - \omega \quad \Longrightarrow \quad \dot{f} = \dot{\theta} - \dot{\omega}. \tag{144}$$

Since

$$\dot{\vec{A}} = \dot{A}\, \hat{A} + A \dot{\hat{A}}, \tag{145}$$

projection onto $\hat{q}$ gives

$$A\dot{\omega} = \dot{\vec{A}} \cdot \hat{q}. \tag{146}$$

Using (130) and (139),

$$A\dot{\omega} = L \left[2F_\theta \sin f - (F_r + \xi F_\theta) \cos f\right]. \tag{147}$$

Therefore

$$\boxed{\dot{\omega} = -\frac{L}{GM\mu^2 e} \left[F_r \cos f + F_\theta \left(\xi \cos f - 2 \sin f\right)\right].} \tag{148}$$

Using

$$\frac{L}{GM\mu^2 e} = \frac{1}{e} \sqrt{\frac{a(1-e^2)}{GM\mu^2}}, \tag{149}$$

we obtain

$$\boxed{\dot{\omega} = -\frac{1}{e} \sqrt{\frac{a(1-e^2)}{GM\mu^2}} \left[F_r \cos f + F_\theta \left(\xi \cos f - 2 \sin f\right)\right].} \tag{150}$$

15.5 Rate of the true anomaly

Since

$$f = \theta - \omega, \tag{151}$$

we have

$$\dot{f} = \dot{\theta} - \dot{\omega}. \tag{152}$$

The instantaneous Keplerian value of $\dot{\theta}$ is

$$\dot{\theta} = \frac{L}{\mu r^2}. \tag{153}$$

Now we apply the same osculating orbit idea and find:

$$\dot{f} = \frac{L}{\mu r^2} - \dot{\omega}\Big|_{\text{osculating}} \tag{154}$$

$$= \sqrt{\frac{GM}{a^3}} \frac{(1 + e\cos f)^2}{(1 - e^2)^{3/2}} - \dot{\omega} \tag{155}$$

where we used

$$r = \frac{a(1 - e^2)}{1 + e\cos f} \tag{156}$$

and

$$L = \mu\sqrt{GMa(1 - e^2)}. \tag{157}$$

15.6 Planar Gauss equations

Collecting the results,

$$\dot{a} = \sqrt{\frac{4a^3}{GM\mu^2(1 - e^2)}} \left[e\sin f\, F_r + (1 + e\cos f)F_\theta\right], \tag{158}$$

$$\dot{e} = \sqrt{\frac{a(1 - e^2)}{GM\mu^2}} \left[F_r \sin f + F_\theta \left(2\cos f + \xi \sin f\right)\right], \tag{159}$$

$$\dot{\omega} = -\frac{1}{e}\sqrt{\frac{a(1 - e^2)}{GM\mu^2}} \left[F_r \cos f + F_\theta \left(\xi\cos f - 2\sin f\right)\right], \tag{160}$$

$$\dot{f} = \sqrt{\frac{GM}{a^3}} \frac{(1 + e\cos f)^2}{(1 - e^2)^{3/2}} - \dot{\omega}, \tag{161}$$

where

$$\xi = \frac{e\sin f}{1 + e\cos f}. \tag{162}$$

For a perturbation confined to the plane,

$$\boxed{i = \dot{\Omega} = 0.} \tag{163}$$

These equations are equations for the *osculating* elements. The quantities a, e, ω are therefore generally functions of time even though the actual position and momentum at each instant are represented by a Keplerian orbit.

16 Orbit-averaged evolution

The element rates above are instantaneous. For a weak perturbation, the long-term evolution is obtained by inserting the unperturbed Keplerian orbit into the right-hand sides and averaging over one orbital period.

For any quantity X,

$$\boxed{\left\langle \dot{X} \right\rangle = \frac{1}{P}\int_0^P \dot{X}\, dt.} \tag{164}$$

For an unperturbed Kepler ellipse,

$$L = \mu r^2 \dot{f}, \tag{165}$$

and therefore

$$\boxed{dt = \frac{\mu r^2}{L}\, df.} \tag{166}$$

To first order in the perturbing force, the distinction between $d\theta$ and df in this averaging relation produces only higher-order corrections.

The average rate can therefore be written as

$$\left\langle \dot{X} \right\rangle = \frac{1}{P}\int_0^{2\pi} \dot{X}\frac{\mu r^2}{L}\, df. \tag{167}$$

A nonzero value of $\left\langle \dot{X} \right\rangle$ represents a secular change. Terms whose integral over $0 \leq f \leq 2\pi$ vanishes describe only short-period oscillations.

17 Nearly circular orbits

As $e \to 0$, the argument of periapsis becomes undefined: a circle has no distinguished periapsis. The explicit $1/e$ in (160) is therefore a coordinate singularity.

Instead of (e, ω), use the two Cartesian components of the eccentricity vector:

$$\boxed{\epsilon_1 = e\cos\omega, \qquad \epsilon_2 = e\sin\omega.} \tag{168}$$

For small e, define

$$\Phi \equiv \theta\big|_{e=0}, \tag{169}$$

and write

$$\theta = \Phi + 2\epsilon_1 \sin\Phi - 2\epsilon_2\cos\Phi + O(e^2). \tag{170}$$

Using

$$r = \frac{a(1-e^2)}{1+e\cos(\theta-\omega)} \tag{171}$$

and retaining terms through first order in ϵ_1, ϵ_2 gives

$$\boxed{r = a\left(1 - \epsilon_1\cos\Phi - \epsilon_2\sin\Phi\right) + O(e^2).} \tag{172}$$

Differentiating,

$$\dot{r} = an\,(\epsilon_1 \sin\Phi - \epsilon_2 \cos\Phi) + O(e^2), \tag{173}$$

where

$$n = \sqrt{\frac{GM}{a^3}}. \tag{174}$$

Also,

$$\boxed{\dot{\theta} = n\,(1 + 2\epsilon_1 \cos\Phi + 2\epsilon_2 \sin\Phi) + O(e^2).} \tag{175}$$

Define

$$\delta\theta \equiv \theta - \Phi = 2\epsilon_1 \sin\Phi - 2\epsilon_2 \cos\Phi. \tag{176}$$

The nonsingular variables ϵ_1, ϵ_2 are preferable to ω near circularity because they remain finite as $e \to 0$.

17.1 First-order equations for nearly circular motion

Substituting the small-e parameterization into the energy equation gives

$$\boxed{\dot{a} = \sqrt{\frac{4a^3}{GM\mu^2}}\,[\delta\theta\, F_r + (1 + \epsilon_1 \cos\Phi + \epsilon_2 \sin\Phi)\, F_\theta] + O(e^2 F).} \tag{177}$$

The eccentricity vector is

$$\vec{A} = GM\mu^2 \left(\epsilon_1 \hat{\Omega} + \epsilon_2 \hat{y}\right) + O(e^2), \tag{178}$$

and its evolution gives

$$\boxed{\dot{\epsilon}_1 = \sqrt{\frac{a}{GM\mu^2}}\,[\sin\Phi\, F_r + 2\cos\Phi\, F_\theta + \delta\theta\,(2\cos\Phi\, F_r - 3\sin\Phi\, F_\theta)] + O(e^2 F),} \tag{179}$$

$$\boxed{\dot{\epsilon}_2 = \sqrt{\frac{a}{GM\mu^2}}\,[-\cos\Phi\, F_r + 2\sin\Phi\, F_\theta + \delta\theta\,(2\sin\Phi\, F_r + 3\cos\Phi\, F_\theta)] + O(e^2 F).} \tag{180}$$

To zeroth order in e,

$$\boxed{\dot{\Phi} = n + O(e^2).} \tag{181}$$

Using Φ as the independent variable,

$$\boxed{\frac{da}{d\Phi} = \frac{2a^3}{GM\mu}\,[\delta\theta\, F_r + (1 + \epsilon_1 \cos\Phi + \epsilon_2 \sin\Phi)\, F_\theta]\,,} \tag{182}$$

$$\boxed{\frac{d\epsilon_1}{d\Phi} = \frac{a^2}{GM\mu}\,[\sin\Phi\, F_r + 2\cos\Phi\, F_\theta + \delta\theta\,(2\cos\Phi\, F_r - 3\sin\Phi\, F_\theta)]\,,} \tag{183}$$

$$\boxed{\frac{d\epsilon_2}{d\Phi} = \frac{a^2}{GM\mu}\,[-\cos\Phi\, F_r + 2\sin\Phi\, F_\theta + \delta\theta\,(2\sin\Phi\, F_r + 3\cos\Phi\, F_\theta)]\,.} \tag{184}$$

18 Worked example: radial inverse-cube perturbation

Consider the perturbing force

$$\boxed{\vec{F} = \frac{\alpha}{r^3}\hat{r}.} \tag{185}$$

Thus

$$F_r = \frac{\alpha}{r^3}, \qquad F_\theta = 0. \tag{186}$$

Because the force is exactly radial,

$$\dot{\vec{L}} = \vec{r} \times \vec{F} = \mathbf{0}. \tag{187}$$

Therefore

$$\boxed{L = \text{constant}, \qquad i = 0, \qquad \dot{\Omega} = 0.} \tag{188}$$

18.1 Semi-major axis

Equation (158) gives

$$\dot{a} = \sqrt{\frac{4a^3}{GM\mu^2(1-e^2)}}\, e\sin f\, \frac{\alpha}{r^3}. \tag{189}$$

Hence

$$\boxed{\dot{a} = \frac{2\alpha a^2}{GM\mu r^3}\dot{r}.} \tag{190}$$

For the Keplerian ellipse introduce the eccentric anomaly u:

$$r = a(1 - e\cos u), \tag{191}$$

and

$$\dot{r} = \frac{ane\sin u}{1 - e\cos u}, \qquad n = \sqrt{\frac{GM}{a^3}}. \tag{192}$$

Therefore

$$\boxed{\dot{a} = \frac{2\alpha en\sin u}{GM\mu(1 - e\cos u)^4}.} \tag{193}$$

Over one unperturbed orbit,

$$dt = \frac{1 - e\cos u}{n}\, du, \tag{194}$$

so

$$\Delta a = \int_0^{2\pi} \dot{a}\, dt \tag{195}$$

$$= \frac{2\alpha e}{GM\mu}\int_0^{2\pi} \frac{\sin u}{(1 - e\cos u)^3}\, du \tag{196}$$

$$= 0. \tag{197}$$

Thus

$$\boxed{\langle \dot{a} \rangle = 0} \tag{198}$$

to first order in the perturbation.

The osculating semi-major axis therefore oscillates during an orbit but has no secular drift.

18.2 Eccentricity

From angular-momentum conservation,

$$L^2 = GM\mu^2 a(1-e^2) \tag{199}$$

and therefore

$$0 = (1-e^2)\dot{a} - 2ae\dot{e}. \tag{200}$$

Hence

$$\boxed{\dot{e} = \frac{1-e^2}{2ae}\dot{a}.} \tag{201}$$

Since $\langle \dot{a} \rangle = 0$,

$$\boxed{\langle \dot{e} \rangle = 0.} \tag{202}$$

18.3 Apsidal precession

For the radial force, (160) reduces to

$$\boxed{\dot{\omega} = -\frac{\alpha L \cos f}{GM\mu^2 e\, r^3}.} \tag{203}$$

To first order in α, evaluate the secular change using the unperturbed Keplerian relation

$$r = \frac{\lambda}{1+e\cos f}, \qquad \lambda = a(1-e^2), \tag{204}$$

together with

$$dt = \frac{\mu r^2}{L}\, df. \tag{205}$$

Then

$$\begin{aligned}
\Delta\omega &= \int_0^{2\pi} \dot{\omega}\, dt && (206)\\
&= -\frac{\alpha}{GM\mu e}\int_0^{2\pi} \frac{\cos f}{r}\, df && (207)\\
&= -\frac{\alpha}{GM\mu e}\int_0^{2\pi} \frac{1+e\cos f}{\lambda}\cos f\, df && (208)\\
&= -\frac{\alpha}{GM\mu e\lambda}\left[\int_0^{2\pi} \cos f\, df + e\int_0^{2\pi}\cos^2 f\, df\right]. && (209)
\end{aligned}$$

Since

$$\int_0^{2\pi} \cos f\, df = 0, \qquad \int_0^{2\pi} \cos^2 f\, df = \pi, \tag{210}$$

we obtain

$$\boxed{\Delta\omega = -\frac{\pi\alpha}{GM\mu\, a(1-e^2)}.} \tag{211}$$

Using

$$P = 2\pi\sqrt{\frac{a^3}{GM}}, \tag{212}$$

the secular rate is

$$\boxed{\langle\dot{\omega}\rangle = -\frac{\alpha}{2\mu\sqrt{GM}\,a^{5/2}(1-e^2)}.} \tag{213}$$

The same result follows directly from the exact Binet equation for the full force

$$F_r = -\frac{GM\mu}{r^2} + \frac{\alpha}{r^3}. \tag{214}$$

Writing $u = 1/r$ gives

$$u'' + \left(1 + \frac{\alpha\mu}{L^2}\right) u = \frac{GM\mu^2}{L^2}, \tag{215}$$

where primes denote derivatives with respect to θ. Hence the radial oscillation has angular frequency

$$\kappa = \sqrt{1 + \frac{\alpha\mu}{L^2}}, \tag{216}$$

and the apsidal angle is

$$\Delta\theta_{\text{apsis}} = \frac{2\pi}{\kappa}. \tag{217}$$

The exact apsidal shift relative to 2π is therefore

$$\Delta\omega_{\text{exact}} = 2\pi\left[\frac{1}{\sqrt{1+\alpha\mu/L^2}} - 1\right]. \tag{218}$$

For a weak perturbation,

$$\frac{\alpha\mu}{L^2} \ll 1, \tag{219}$$

so

$$\Delta\omega_{\text{exact}} = -\frac{\pi\alpha\mu}{L^2} + O(\alpha^2) \tag{220}$$

$$= -\frac{\pi\alpha}{GM\mu\,a(1-e^2)} + O(\alpha^2), \tag{221}$$

which agrees with the orbit-averaged planetary equation.

19 Rømer delay for a Keplerian binary

For a binary of component masses m_1 and m_2,

$$M = m_1 + m_2. \tag{222}$$

Let $\vec{r}$ be the relative separation and let the primary's barycentric position be

$$\vec{x}_1 = \frac{m_2}{M}\vec{r}. \tag{223}$$

If $\hat{R}$ points from the source towards the observer, the light-travel-time contribution is

$$\Delta_R = -\frac{\vec{x}_1 \cdot \hat{R}}{c}. \tag{224}$$

Using the orbital-plane basis,

$$\vec{r} = r\left[\cos\theta\,\hat{\Omega} + \sin\theta\,\hat{y}\right], \tag{225}$$

with

$$\theta = \omega + f, \tag{226}$$

and

$$\hat{y}\cdot\hat{R} = \sin\iota, \qquad \hat{\Omega}\cdot\hat{R} = 0, \tag{227}$$

we obtain

$$\Delta_R = \frac{m_2}{Mc}\,r\sin\iota\,\sin(\omega+f). \tag{228}$$

Define

$$a_1 = \frac{m_2}{M}a, \tag{229}$$

and use the eccentric-anomaly relations

$$r\cos f = a(\cos u - e), \tag{230}$$

$$r\sin f = a\sqrt{1-e^2}\sin u. \tag{231}$$

Since

$$\sin(\omega+f) = \sin\omega\cos f + \cos\omega\sin f, \tag{232}$$

we obtain

$$\boxed{\Delta_R = \frac{a_1\sin\iota}{c}\left[(\cos u - e)\sin\omega + \sqrt{1-e^2}\sin u\cos\omega\right].} \tag{233}$$